migrating your data to a filesystem with Max I/O mode, you can increase the I/O limit from 35,000 IOPS to 500,000+ IOPS. Keep in mind, however, that doing so increases read latency from around 600 microseconds to single-digit milliseconds. The next listing shows the CloudWatch alarm to monitor the `PercentIOLimit` metric.

Listing 9.6 Monitoring the `PercentIOLimit` metric using a CloudWatch alarm

```
PercentIOLimitTooHighAlarm:
  Type: 'AWS::CloudWatch::Alarm'                          ①
  Properties:
    AlarmDescription: 'I/O limit has been reached, consider ...'
    Namespace: 'AWS/EFS'                                  ②
    MetricName: PercentIOLimit                            ③
    Statistic: Maximum                                    ④
    Period: 600                                           ⑤
    EvaluationPeriods: 3                                  ⑥
    ComparisonOperator: GreaterThanThreshold              ⑦
    Threshold: 95                                         ⑧
    Dimensions:                                           ⑨
    - Name: FileSystemId                                  ⑩
      Value: !Ref FileSystem
```

① Creates a CloudWatch alarm

② EFS uses the AWS/EFS for all its metrics.

③ If the PercentIOLimit metric reaches 100%, we are hitting the limit of 35,000 read or 7,000 write operations. Switching to the Max I/O performance mode is a possible mitigation.

④ The alarm evaluates the PercentIOLimit metric by using the maximum.

⑤ The alarm monitors a period of 600 seconds.

⑥ The alarm takes into consideration the last three periods, which is 3x 600 seconds.

⑦ The alarm will trigger if the maximum PercentIOLimit was greater than the threshold.

⑧ The threshold is set to 95%.

⑨ Each filesystem reports its own PercentIOLimit metric. Therefore, we are configuring a dimension here.

⑩ The dimension is named FileSystemId.

References the ID of the filesystem